

PQC Adoption Gap in 2026: Browser Readiness vs. Server-Side Lag

Day 16 | Category: PQC Adoption & Industry Status

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PQC Adoption Gap in 2026: Browser Readiness vs. Server-Side Lag The transition to Post-Quantum Cryptography (PQC) is a critical step in securing our digital infrastructure against the threat of quantum computers. However, as of 2026, the adoption of PQC is not uniform. While major web browsers have made significant strides in supporting PQC, server-side adoption is lagging behind, creating a significant gap that needs to be addressed. ### The "Harvest Now, Decrypt Later" Threat The urgency to adopt PQC is driven by the "Harvest Now, Decrypt Later" (HNDL) threat model. This is where adversaries record encrypted traffic today, with the intention of decrypting it in the future when quantum computers become powerful enough to break current encryption algorithms. This makes the adoption of PQC a pressing issue, even if large-scale quantum computers are not yet a reality. ### Browser Support: Leading the Way All major web browsers now support hybrid post-quantum key exchange by default. This means that they use a combination of a classical algorithm (like X25519) and a post-quantum algorithm (like ML-KEM/Kyber768). This hybrid approach ensures that the connection remains secure even if one of the algorithms is compromised.

Browser	PQC Key Exchange	Default Enabled	Algorithm
Chrome 124+	✓	✓	X25519Kyber768
Firefox 128+	✓	✓	X25519Kyber768
Safari 18+	✓	✓	X25519Kyber768
Edge 124+	✓	✓	X25519Kyber768

Server-Side Support: The Lagging Follower While browsers are ready for PQC, the server-side is a different story. The adoption rate of PQC on web servers is significantly lower. While a considerable percentage of top websites have enabled hybrid PQC, the vast majority of the internet is still not PQC-ready.

Adoption Statistics (2026 Estimates)

- Top 100 websites: ~40% support hybrid PQC
- Top 1,000 websites: ~25% support hybrid PQC
- Top 1M websites: ~8.6% support hybrid PQC
- Enterprise internal services: < 2%

This gap between browser readiness and server-side implementation is a major hurdle in the transition to a quantum-resistant internet. ### Daily Quiz 1. What is the "Harvest Now, Decrypt Later" (HNDL) threat? 2. What is the hybrid approach to PQC key exchange? 3. What is the main challenge in the widespread adoption of PQC?